

 **คณะแพทยศาสตร์ศิริราชพยาบาล มหาวิทยาลัยมหิดล**
FACULTY OF MEDICINE SIRIRAJ HOSPITAL

Lower Extremity Injuries

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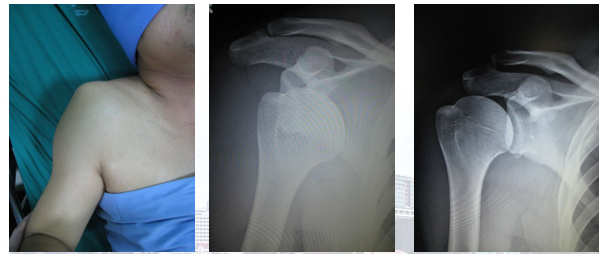
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Fracture



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Dislocation



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Outlines

- Mechanism of injury
- Principle management
- Complications
- Lower extremity injury



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Mechanism of Injury

- High/Low energy



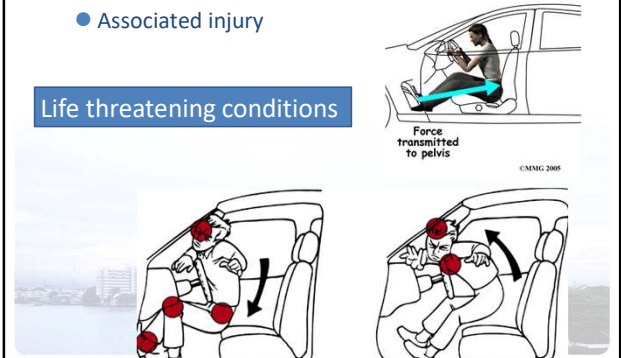
Colles fracture

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Mechanism of injury

- Associated injury

Life threatening conditions

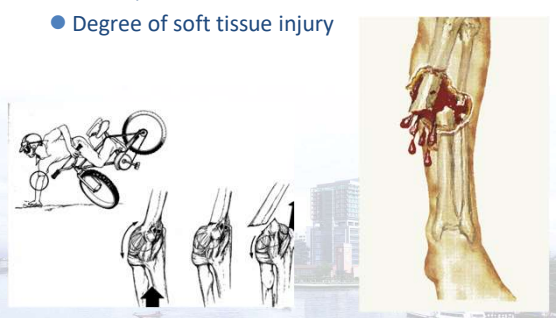


Force transmitted to pelvis

© MMG 2008

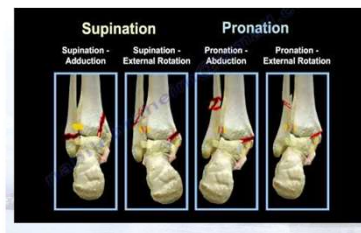
Mechanism of injury

- Direct / Indirect force
- Degree of soft tissue injury



Mechanism of injury

- Direction of force
- Fracture configuration

Principle of management

- Recognition
 - ATLS
 - Hx, PE
 - Radiograph
- Reduction
 - Close technique
 - Open technique
- Retention
 - External support/fixation
 - Internal fixation
- Rehabilitation



Recognition

- Primary survey
 - Immediate life threatening conditions
- Secondary survey
 - Head to toe evaluation
 - Associated injury
 - Neurovascular injury
 - Missed injury



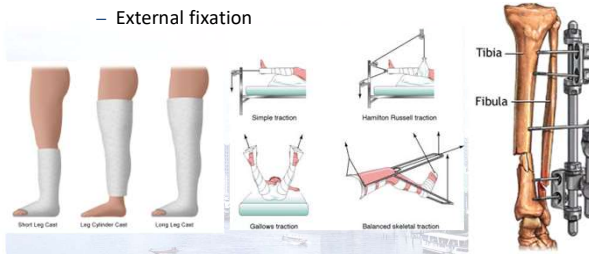
Reduction

- Closed reduction
- Open reduction



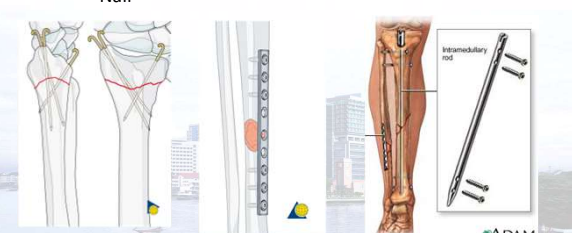
Retention

- External
 - Cast or slab
 - Traction
 - External fixation



Retention

- Internal
 - K-wire
 - Plate and screw
 - Nail



The diagrams show three methods of internal fixation: 1. K-wire: A thin wire inserted into the bone. 2. Plate and screw: A metal plate with screws attached to the bone. 3. Intramedullary nail: A long nail inserted into the medullary canal of the bone. The background shows a cityscape with a river and boats.

Rehabilitation

- Range of motion exercise
- Stretching exercise
- Strengthening exercise
- Etc.



The background image shows a cityscape with a river and boats, likely representing a hospital or medical facility.

Open fracture

- Associated injury
- Management
- Timing



The images show three types of open fractures: 1. A small laceration on the leg. 2. A larger laceration with exposed bone. 3. A severe open fracture with significant bone loss and soft tissue damage. The background shows a cityscape with a river and boats.

Gustilo-Anderson classification

Gustilo classification

Type	Wound	Level of Contamination	Soft Tissue Injury	Bone Injury
I				
II				
III ^A				
A				
B				
C				

Converting factors

- Contamination
 - Soil
 - Dirty water
 - Bites
 - Barnyard
 - Gross appearance
- More than 12 hours
- High energy injury
 - Bone loss
 - Segmental fracture
 - Crush injury
 - Degloving injury
 - Compartment syndrome
 - Flap coverage require



The background image shows a cityscape with a river and boats, likely representing a hospital or medical facility.

Open fracture

- Timing
 - Not more than 24 hours
- Antibiotic
 - Type I,II: 1st gen cephalosporin
 - Type III: 1st gen cephalosporin + aminoglycoside
 - Penicillin 2million unit q 4 hr in soil contamination



The background image shows a cityscape with a river and boats, likely representing a hospital or medical facility.

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- Please give the factors affecting the Gustilo classification?



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Complications

- Vascular injury
- Compartment syndrome
- Thromboembolism
- Fat embolism syndrome



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Vascular injury

- Arterial injury
 - Intimal flap
 - Disruptions or subintimal hematoma
 - Wall defect
 - Complete transection
 - A-V fistula
- Venous injury
- Upper extremity has more extensive collateral circulation

Blunt

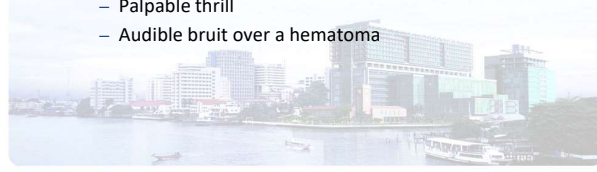
Penetrating



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Vascular injury

- Hard signs
 - Classic signs of arterial occlusion (6P)
 - Pulselessness, Pallor, Paresthesia, Pain, Paralysis, Poikilothermia
 - Massive bleeding
 - Rapidly expanding hematoma
 - Palpable thrill
 - Audible bruit over a hematoma



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Vascular injury

- Soft signs
 - Hx of arterial bleeding at the scene
 - Proximity of wound to an artery
 - Small, non-pulsatile hematoma over artery extremity
 - Neurologic deficit originating in a nerve adjacent to a named artery

Awareness **Frequent exam** **Investigation**





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Vascular injury

- Management
 - If limb threatening
 - Immediate exploration
 - If localization is necessary
 - Intraoperative angiography
 - Duplex U/S





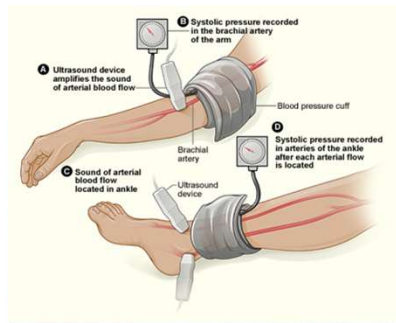

Vascular injury

- Arterial Pressure Index (API)
 - Doppler pressure in the injured extremity
the uninjured extremity
- Ankle Brachial Index (ABI)
 - Doppler pressure in the injured extremity
the brachial artery

Doppler ultrasound



Ankle brachial index



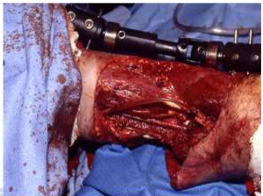
> 0.9

Vascular injury

- Orthopaedic management
 - Hemorrhage control
 - Finger compression
 - Pressure dressing
 - Blood pressure cuff
 - Realignment and splinting
 - only kinking not spasm
 - Measured ABI
 - Angiography
 - ABI < 0.9
 - Cannot be obtained due to distal injury
 - Difficult to evaluate: children

Orthopaedic and vascular injury

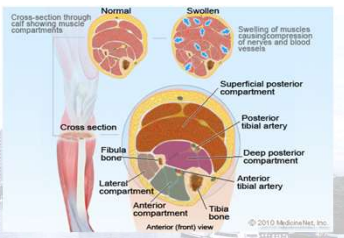
- Priority: Who first?
 - Stable orthopaedic injury
 - Ischemic time
- Stable injury
 - Vascular first
- Unstable injury
 - Orthopaedic first
- Unstable and delayed injury
 - Temporary shunting



- Please provide the hard signs of vascular injury ?

Compartment syndrome

- An **increased pressure** within an **enclosed osteofascial space** that reduces the capillary blood perfusion below a level necessary for tissue viability



Compartment syndrome

- **Causes**
 - Fracture
 - Soft tissue injury
 - Arterial injury
 - Limb compression
 - Burns



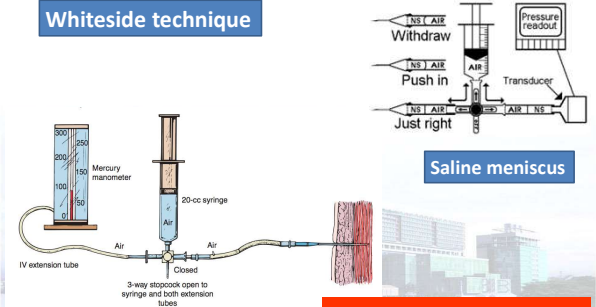
Compartment syndrome

- **Symptoms**
 - Pain out of proportion
 - Pain is unrelenting
 - No relief following splinting or removal of casts & bandages
 - Paresthesia
- **Signs**
 - Pain on palpation of compartment
 - Tense / swollen compartment
 - Passive muscle stretch → severe pain
 - Sensory deficit of nerve in the compartment
 - Muscle weakness



Compartment syndrome

Whiteside technique



Normal pressure is 0-8 mmHg

Saline meniscus

Transducer

Pressure reading

Withdraw

Push in

Just right

Mercury manometer

20-cc syringe

IV extension tube

3-way stopcock open to syringe and both extension tubes

Closed

Compartment syndrome

Commercial set



Compartment syndrome

- **Magic number**
 - 30 mmHg (corresponds with normal capillary pressure)
 - 45 mmHg (capillary pressure rises in compartment syndrome)
 - 20 mmHg below DBP
 - 30 mmHg below DBP

Mubarak, SJ & Hargens, AR

Matsen, FA

Whiteside, TE

McQueen, MM & Court-Brown, CM



Compartment syndrome

Clinical Impression: Compartment Syndrome

Unconscious or unreliable patients
Inconclusive clinical findings

Head Injured Patient
Unresponsive
Polytrauma
Questionable Clinical Findings

Measurement of Compartment Pressures
Determine Blood Pressure
Diastolic and Mean Arterial

Definitive Clinical Findings

$\Delta P \leq 40$ mmHg

$\Delta P > 40$ mmHg

Continuous compartment Pressure monitor and serial clinical evaluation

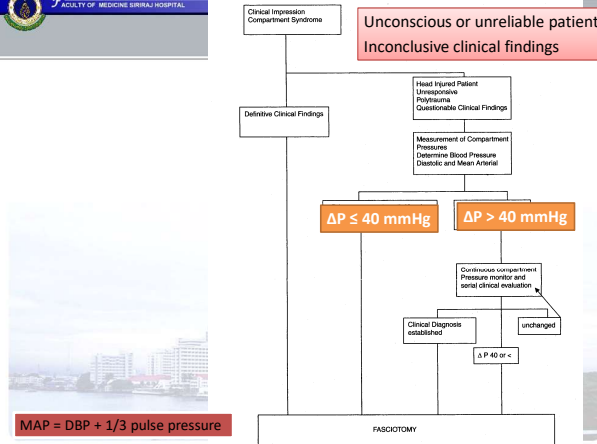
Clinical Diagnosis established

unchanged

$\Delta P 40$ or $\leq$

FASCIOTOMY

MAP = DBP + 1/3 pulse pressure



Compartment syndrome

- Release constrictive dressings
- Bivalve cast and webril
- Fasciotomy
- Fracture stabilization

Compartment syndrome

- What is the clinical presentation of compartment syndrome?
- When we should do compartment pressure monitoring ?

Thromboembolism

- Risk factors
 - Hx of DVT/PE
 - Hx of thromboembolic or malignancy
 - Hemiplegia/ hemiparesis
 - Acute spinal cord injury
 - Elective major lower extremities injury
 - Pelvic, hip, knee injury
 - BMI > 30 kg/m²
 - Congestive heart failure

Thromboembolism

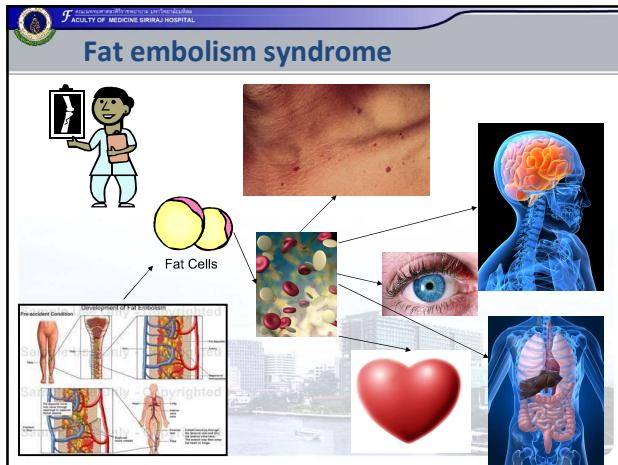
- Clinical findings
 - Calf and thigh swelling
 - Discoloration
 - Warmth
 - Pain along deep venous system
 - Unilateral pitting edema
 - Homans' sign
 - Forceful abrupt dorsiflexion
 - Pain in calf

Thromboembolism

- Diagnosis
 - Venography
 - Duplex ultrasound
- Management
 - Anticoagulant
 - Consider prophylaxis

Fat embolism

- Fat emboli
 - 90% after major trauma
- Fat embolism syndrome
 - 2-5%
 - 24-72 hours after injury
 - Risk factor



Fat embolism syndrome	
Major criteria	Petechial rash
	Respiratory insufficiency
	Cerebral involvement
Minor criteria	Tachycardia
	Fever
	Retinal changes
	Jaundice
	Renal signs
	Thrombocytopenia
	Anaemia
	High ESR
	Fat macroglobinemia

2 major or 1 major + 4 minor criteria

Gurd JBJS Br 1970

Risk factor

- Males
- Age 10-39 years
- Post traumatic hypovolemic state
- Reduce cardiopulmonary reserve

General

- Intramedullary reamed and unreamed nailing after femoral fractures
- Joint replacement after femoral fractures
- Bilateral procedures
- Joint replacement with high-volume prosthesis

Surgery

- Multiple fractures
- Bilateral femur fractures
- Femoral shaft fractures
- Lower extremity fractures
- Traumatic fractures
- Concomitant pulmonary injury

Injury

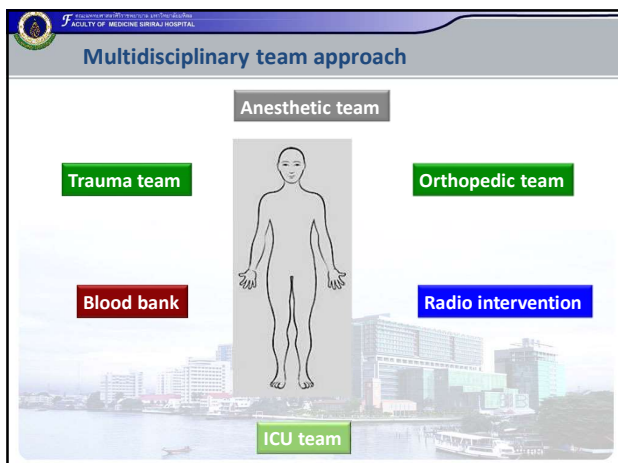
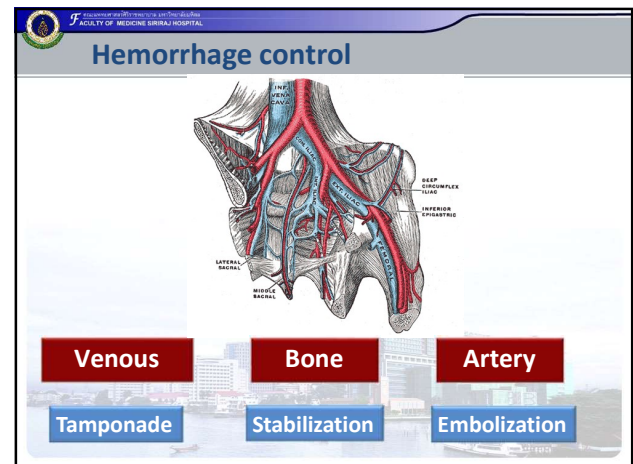
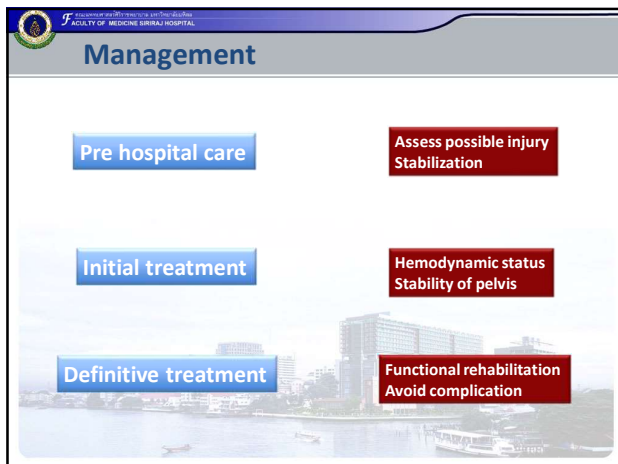
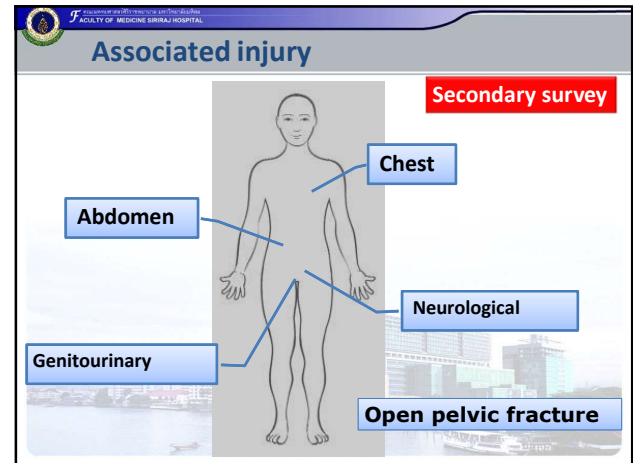
Fat embolism syndrome

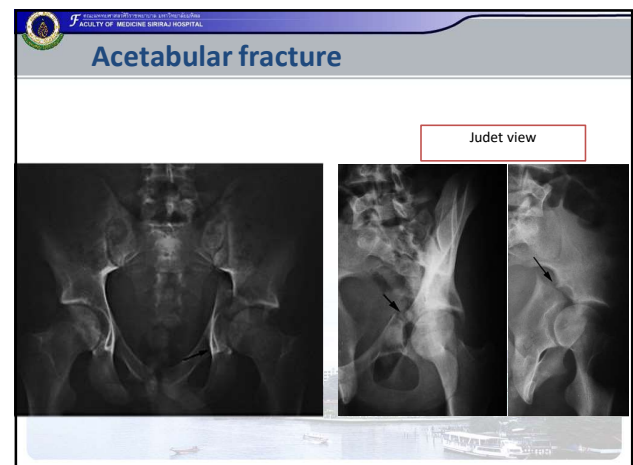
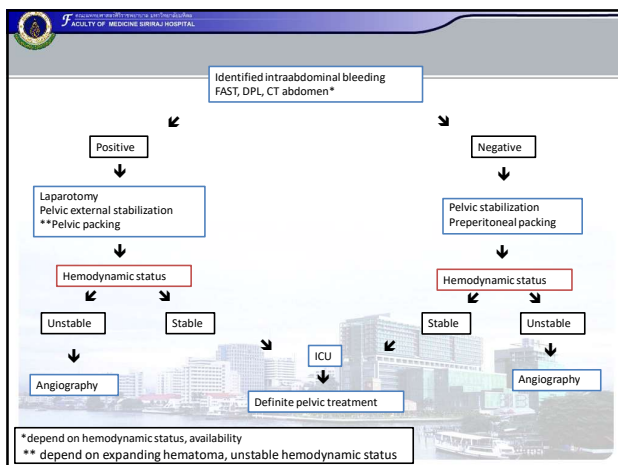
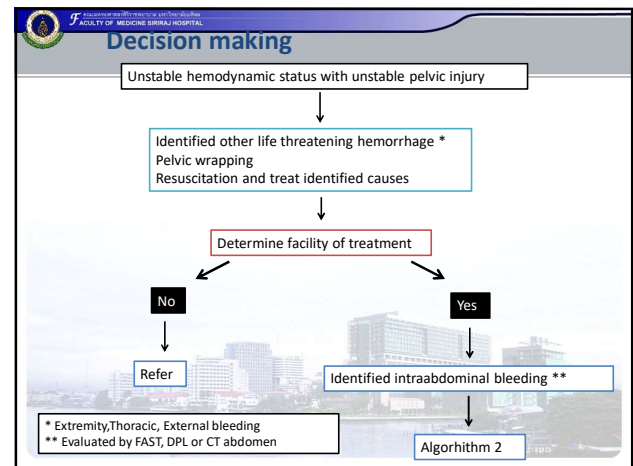
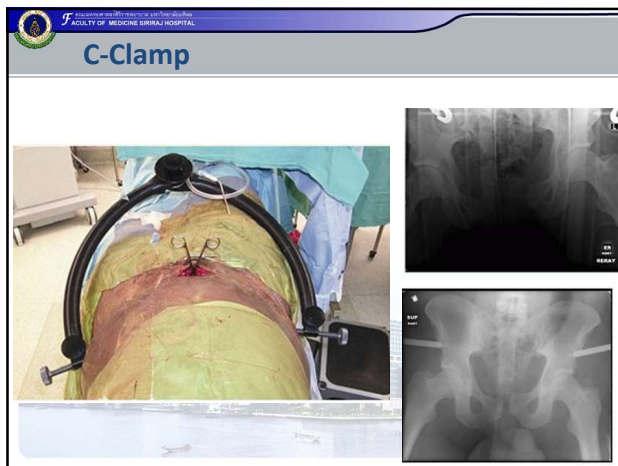
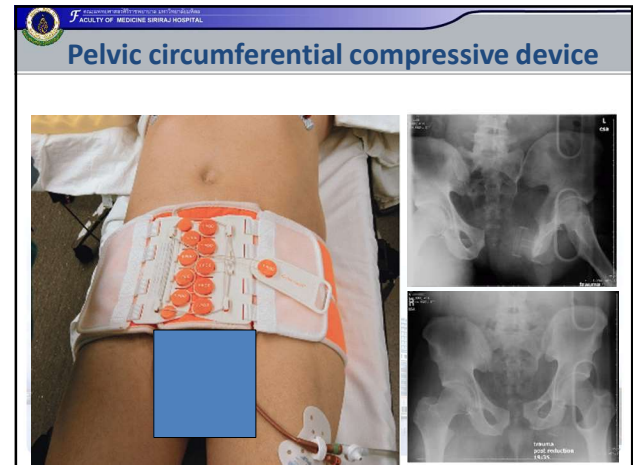
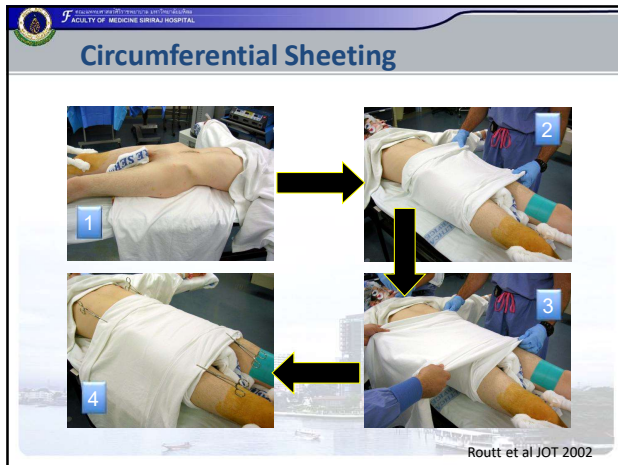
- **Management**
 - Supportive treatment
 - Intensive monitoring
 - Early fracture stabilization

• Please provide major criteria for diagnosis fat embolism syndrome

Lower extremity injury

- Pelvis
- Hip fracture/dislocation
- Femur
- Knee
- Tibia
- Ankle
- Foot





Hip dislocation

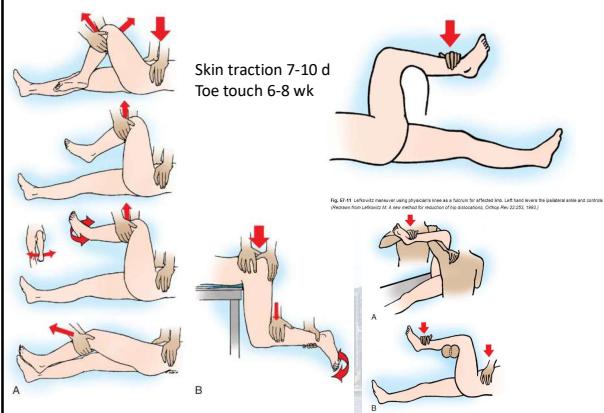


Position
Adduction, Flexion, IR

X-rays
Overlap roof acetabulum
Smaller head
Post. wall fx

Hip dislocation

- Emergency reduction < 6 hr
 - Under GA
 - Incongruent is not Emergency
- Post reduction
 - Film – fracture, congruent
 - Sciatic n. test



Skin traction 7-10 d
Toe touch 6-8 wk

Fig. 67-40 A, Allis maneuver; B, Simon maneuver.

Fig. 67-41 Left hip dislocation using physician's knee as a fulcrum for affected hip. Left hand wears the posterior wedge and controls rotation. (Reprinted from Campbell et al. in: Principles and Practice of the Hip, 2nd ed., 1984.)

Fig. 67-42 A, Distraction of the hip with traction and assistant's arms as fulcrum and wedge as lever. B, A second assistant stabilizes femoral head.

Hip fracture

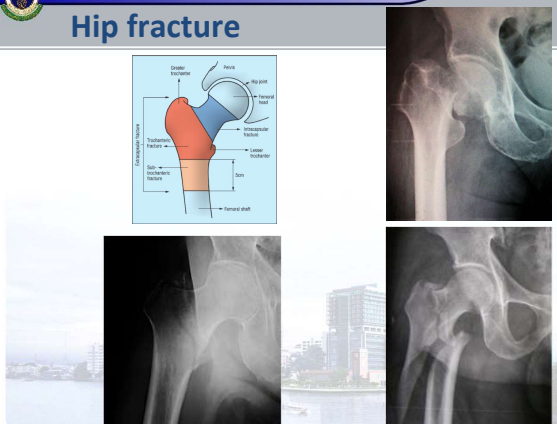


Diagram labels: Greater trochanter, Lesser trochanter, Femoral head, Femoral neck, Femoral shaft, Distal femur, Proximal femur, Femoral head, Femoral neck, Femoral shaft, Distal femur.

Hip fracture

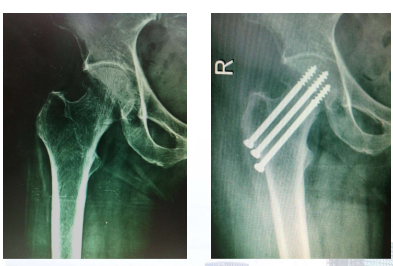
Morbidity

- Pressure sore
- Pneumonia
- Urinary tract infection
- Venous thromboembolism

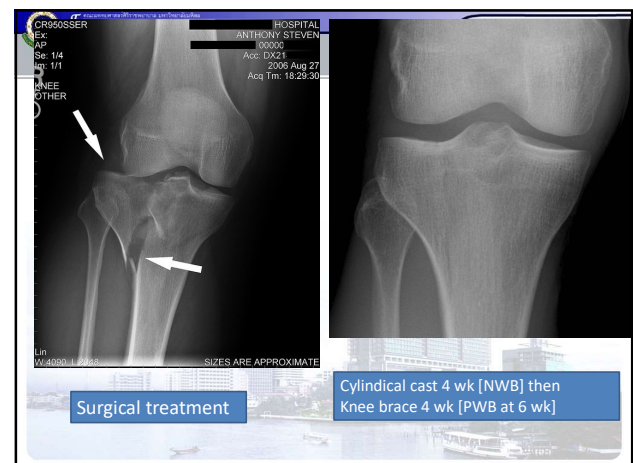
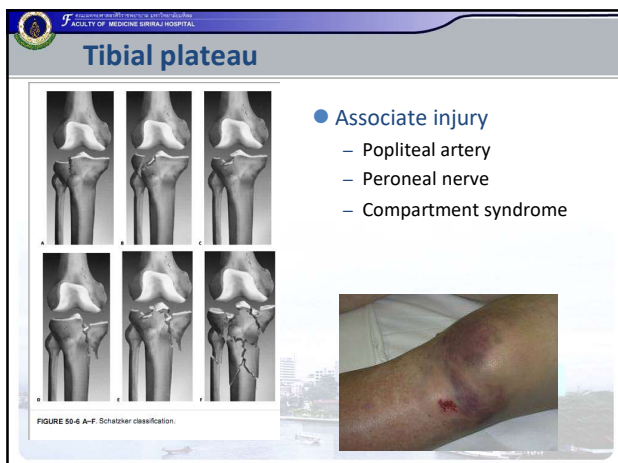
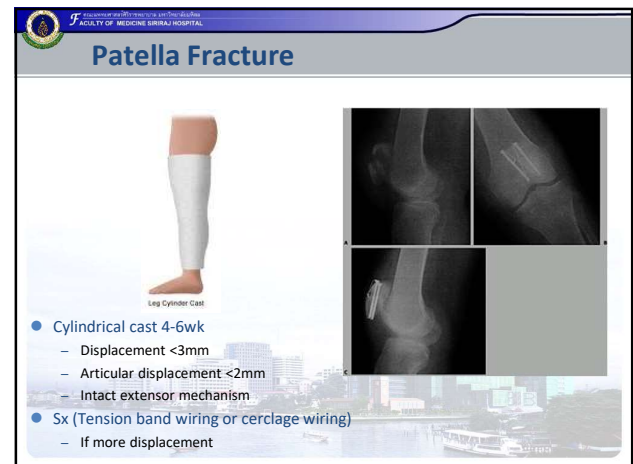
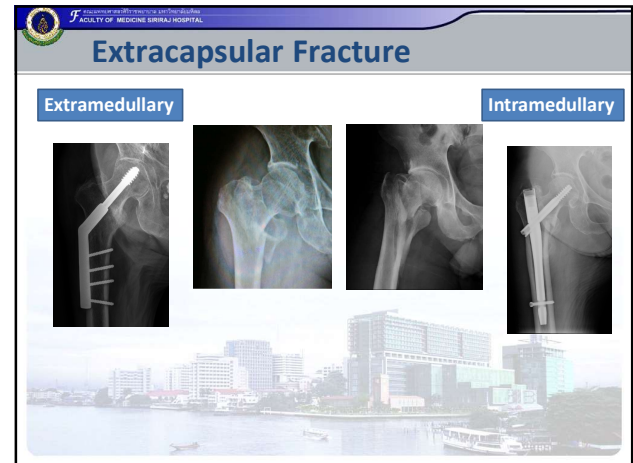
Goal: early ambulation



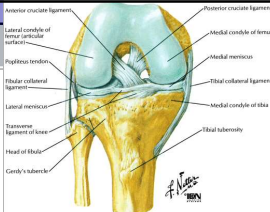
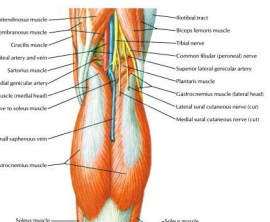
Non Displaced Intracapsular Fracture



Fixation in situ



Knee dislocation

Neurovascular injury

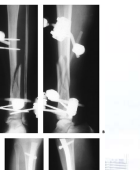
Shaft tibia




- **Reduced Simple, short oblique, spiral Fx**
 - Long leg cast 4-6 wk [NWB]
 - [form callus] → PTB 4-6 wk [PWB]
- **Acceptable Fracture Reduction**
 - < 5 degrees - Coronal plane
 - < 10 degrees - Sagittal plane
 - < 10 degrees - rotational
 - < 1 cm - shortening
 - 5 mm of distraction may delay healing 8 to 12 months.
 - >50% cortical contact





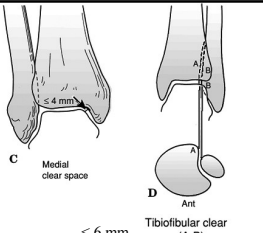


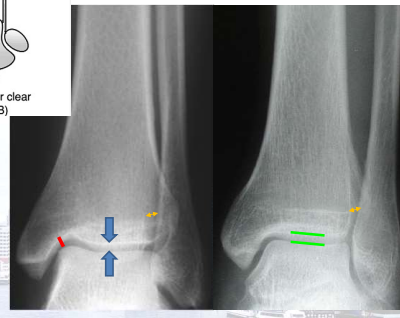

- **Indication for surgery**
 - Complex Fx
 - Open fx
 - Vascular injury
 - Irreducible
 - Severe soft tissue injury
 - Multiple Fx
 - Delay/nonunion

Ankle Fracture



- **Sprain VS Fracture**
- **Ottawa Ankle Rules**
 - Bone tenderness along 6 cm from distal fibular
 - Bone tenderness along 6 cm from distal tibia
 - Inability to bear weight 4 steps



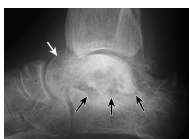


AP Mortise

Talar neck fracture

Almost surgery





High rate of AVN



Calcaneus

Normal

- **Bohler angle** 20-40°
- **Gissane's angle** 105 - 135°

- Extra articular
- Intra articular
 - Tongue type
 - Joint depression
- Rx
 - Displacement
 - Surgery
 - Nondisplacement
 - Cast 2-3 Mo [NWB]




Lateral **Axial**




Film

- Lateral calcaneus
- Axial
- Broden
- Foot AP



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Thank You

